

ORIGINAL ARTICLE

Brevity is the Soul of Twitter: The Constraint Affordance and Political Discussion

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Many hoped that social networking sites would allow for the open exchange of information and a revival of the public sphere. Unfortunately, conversations on social media are often toxic and not conducive to healthy political discussions. Twitter, the most widely used social network for political discussions, doubled the limit of characters in a tweet in November 2017, which provided an opportunity to study the effect of technological affordances on political discussions using a discontinuous time series design. Using supervised and unsupervised natural language processing methods, we analyzed 358,242 tweet replies to U.S. politicians from January 2017 to March 2018. We show that doubling the permissible length of a tweet led to less uncivil, more polite, and more constructive discussions online. However, the declining trend in the empathy and respectfulness of these tweets raises concerns about the implications of the changing norms for the quality of political deliberation.

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Over the past decade, we have witnessed a shift in political communication. In the context of political campaigning, social media platforms such as Twitter and Facebook now provide politicians with a platform to mobilize and engage in meaningful discussions with their constituents (Theocharis, Barberá, Fazekas, Popa, & Parnet, 2016). At the same time, these platforms have given citizens the ability to access political information, participate in political movements, and voice their support for or dissent against their government. Social networking sites are now essential tools in the democratic and civic process, as they offer a neutral and open platform for dialogue, act as the bedrock where citizens build ideas about politics through everyday talk (Wojcieszak & Mutz, 2009; Wyatt, Katz, & Kim, 2000), and

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facilitate direct contact between citizens and their political representatives (e.g., Liu & Zhang, 2013; Tufekci & Wilson, 2012).

Everyday political discussions on social networking sites are considered to be one of the many pathways to political participation (for a detailed review, see Cook, Carpini, & Jacobs, 2007; Eveland, Morey, & Hutchens, 2011; Hopmann, Matthes, & Nir, 2015). However, the question of whether meaningful political discussions are possible on social networking sites depends on whether or not it is possible to have rational discussions (Habermas, 1984) and to communicate ideas and build consensus in a polite and respectful manner (e.g., Liu & Zhang, 2013; Mutz & Reeves, 2005; Papacharissi, 2004). Some researchers are skeptical of the potential of social networking sites as public spheres for political deliberation (Bail et al., 2018; Mendelberg, 2002; Schkade, Sunstein, & Kahneman, 2000). One of the concerns raised is that political discussions on platforms such as Twitter are of low quality, since they are often toxic, provocative, and replete with trolls (e.g., Berry & Sobieraj, 2013; Nithyanand, Schaffner, & Gill, 2017; Theocharis et al., 2016).

Many explanations for the low quality of online discourse have been put forth, including the existence of disinformation campaigns, the polarization of politics, spurious accounts, and echo chambers (e.g., Berry & Sobieraj, 2013; Chen, 2017). Incivility ultimately limits users' ability to have meaningful and constructive exchanges with each other or with their political representatives (Theocharis et al., 2016).

We argue that another factor that regulates the quality of political discussions are the technological affordances for communication. Affordances are properties that enable or constrain the potential for action (Faraj & Azad, 2012). Examples of affordances in digital communication include the ability to have real-time discussions, to hide one's true identity, and to navigate information in a hierarchical manner (Friess & Eilders, 2015; Sundar, 2008). A number of studies have shown an association of one affordance (anonymity) with the quality of online political discussions (e.g., Halpern & Gibbs, 2013; Theocharis, Lowe, van Deth, & García-Albacete, 2015; Towne & Herbsleb, 2012).

In this paper, we examine whether another affordance—constraints on message length—impacts the quality of online political discussions. Using an interrupted times series design, combined with an automated content analysis of over 350,000 tweets related to politics, we compared various message features associated with ideal political discussions before and after Twitter moved from allowing 140 characters to 280 characters per tweet. We show that Twitter's relaxation of its message length constraint affected some, but not all, message features, which improved the quality of political discussions. For instance, while the changeover increased the prevalence of less uncivil, more formal, and more constructive messages, there was a decrease in the empathy and respectfulness of messages. We also tested the robustness of our results using different model specifications and bandwidths, and a placebo test.

Our theoretical contribution is two-fold. First, we extend prior research by shifting attention to social media affordances that ultimately influence the quality of discussions on social media. Changes to the design of platforms, which are more

tractable than, for instance, censoring content, have substantive effects on the health of online political discussions. Second, by revealing the trade-offs in content and style that users face when articulating their opinion, our study yields valuable insights about how the use of these platforms may affect the quality of public discourse and users' experiences of online political participation as a more reactive versus a more deliberative exercise. We also offer a methodological contribution in the form of language models that can automatically label text according to its uncivil and deliberative qualities.

The Twittersphere

Political discussions on Twitter and other online platforms allow people with diverse perspectives and opinions to participate in casual conversations. In theory, they can cut across diverse social networks. In addition to facilitating horizontal discussions between citizens, platforms like Twitter allow vertical discussions between citizens and policymakers (e.g., Ausserhofer & Maireder, 2013; Davis, 2010; Effing, van Hillegersberg, & Huibers, 2011).

Twitter's users organically create a networked sphere of political discussions that is structurally independent from the traditional arena of politics or news; yet, it connects with the two through official affiliations and real-life interactions (Lindgren & Lundström, 2011). The online political sphere has the potential to reinvigorate offline politics by allowing millions of individual contributions, thus subverting the often-monolithic agenda set by traditional mass media and policymakers in the offline world (e.g., Habermas, Lennox, & Lennox, 1974; Papacharissi, 2010; Shirky, 2008).

The fact that Twitter facilitates discussions between citizens and policymakers does not, in itself, make it a "democratic utopia" (Papacharissi, 2004; Stroud, Scacco, Muddiman, & Curry, 2015). Online political discussions rarely meet the ideals for political deliberation, such as open communication, equality, inclusivity, and compromise (e.g., Friess & Eilders, 2015; Nithyanand et al., 2017; Theocharis et al., 2016; Wojcieszak, 2010). Political discussions on Twitter are often replete with toxic and abusive responses, and group-based stereotyping (Halpern & Gibbs, 2013; Theocharis et al., 2016). Users participating in discussions on Twitter are found to be unlikely to indulge in reflection or frame coherent arguments, which negatively impacts the quality of political discussions (Janssen & Kies, 2005; Stromer-Galley & Martinson, 2009). A study by Theocharis et al. (2016) argued that users' toxic and uncivil responses were responsible for shutting down any meaningful political engagement between elected U.S. politicians and the citizenry.

To summarize, many studies have identified a gap in the normative ideals of Twitter as an ideal platform for lively and inclusive political debates and its role in actually facilitating political deliberation. The following section contextualizes the study's hypotheses in the current understanding of how Twitter's technological features, or affordances, enable deliberation and politeness or constrain incivility.

Technological affordances and the potential for civic discussions

When designing online platforms, developers make many choices that change “possibilities for action between an object/technology and the user that enables or constrains potential behavioral outcomes in a particular context” (Evans, Pearce, Vitak, & Treem, 2017, p. 36). These possibilities for action are often called affordances (Gibson, 1977; Greeno, 1994) and can be broadened or narrowed by changing the underlying technical specifications. This study corroborates the perspective that, rather than studying social media platforms as a single entity, a quantitative analysis should instead operationalize social media platforms as a mixture of technological affordances that together define the properties of the platform, such as its interactivity, synchronicity, organization of members and content, and control given to the user (Eveland, 2003; Walther, Gay, & Hancock, 2005). For instance, in the context of political discussions, one of the ways Twitter affords the scope for two-way communication between politicians and citizens is through the way asymmetrical friendships are specified by default (Grant, Moon, & Busby Grant, 2010). The affordance for two-way communication is restricted on Facebook, where, in the default setting, two individual users may need to “follow” each other in order to send a message or even view each other’s profile. Similarly, we can consider other attributes of Twitter’s design which afford or inhibit political discussions.

Friess and Eilders (2015) and Janssen and Kies (2005) identified some affordances that affect political deliberation. Anonymity, for instance, affords higher participation but uncivil discourse (Towne & Herbsleb, 2012). Real-time participation in synchronous chats provokes instantaneous reactions but reduces reflexive, coherent argumentation and rebuttal (Janssen & Kies, 2005; Stromer-Galley & Martinson, 2009). Flattened follower-followee connections overcome the depersonalizing effects of digital communication and increase citizens’ emotional closeness to political elites and elected candidates (Lee & Oh, 2012).

Navigable information interfaces support political deliberation by fostering clear communication, rational argumentation, and constructiveness (Towne & Herbsleb, 2012). Among the studies exploring the role of affordances in political deliberation, a majority have focused on how anonymity and deindividuation reduce the rationality, sincerity, and civility of the conversation (e.g., Halpern & Gibbs, 2013; Theocharis et al., 2015; Towne & Herbsleb, 2012). Anonymity also improves the likelihood of participation and, hence, the quantity and inclusivity of political debates (Towne & Herbsleb, 2012).

At least two studies have examined the relationship between message length (another affordance, and our particular focus) and political discussion quality. However, these studies turned up conflicting findings: on the one hand, Papacharissi (2004) reported that longer messages posted in political discussions were significantly more uncivil than shorter messages. Other publications have argued that Twitter’s character limit constraints can encourage individual creativity and improve content quality (Gligorić, Anderson, & West, 2018; Joyce, 2009). On the other hand, Oz, Zheng, and Chen (2018) found that shorter messages on Twitter were more uncivil,

more impolite, and less deliberative than longer messages on Facebook; however, in their experimental replication, they observed only a significant improvement in deliberation in the longer messages.

These studies offer important insights; however, they are limited in crucial ways. For instance, Papacharissi (2004) did not account for self-selection and the possibility that those that tend to write longer messages may already behave more uncivilly. While the Oz et al. (2018) experiment solves this internal validity issue, the character limit is confounded with other characteristics and norms of the platform. That is, subjects responded to either a mock Facebook post or a mock Twitter post. Given these conflicting findings, we began with a research question:

RQ1: What effect does doubling the character limit have on incivility?

Character limits may also limit the formality of discourse, making it less amenable to clear and open communication. As the original Twitter users embraced its character limit, they organically devised a series of conventions to convey meaning and information structure: slang, netspeak, and abbreviations were adopted to express reactions (LOL, SMH), share opinions (IMHO, AFAIK), reference other users (-mentions, h/t, via), label topics (hashtags), and identify propagated messages (RT, QT). However, while the need for concision required adherence to convention, this may have led to the sacrifice of clarity over the content. Twitter users were constrained to expressing themselves in 140 characters, so they were more likely to use slang and abbreviations to compress their ideas, as compared to their behavior on platforms such as Facebook, which do not have such constraints (Jaidka, Guntuku, Buffone, Schwartz, & Ungar, 2018; Lin & Qiu, 2013). With the availability of more characters, however, users are more likely to apply a less casual and more professional writing style, as has been reported in some previous studies (Gligorić et al., 2018; Jaidka et al., 2018). Accordingly, we posited the following hypothesis:

H1: Comments posted on Twitter after the 280-character limit change will be less informal (more polite) than before the change.

Civility is not a sufficient condition for achieving the democratic potential of the Internet. Political discussions also need to be substantive, in terms of providing evidence for their claims and raising arguments that aim to build consensus or move the conversation forward. However, Papacharissi (2004) raised the concern that greater adherence to politeness could restrict political debate by substituting it with a formal exchange of rhetoric. With fewer stakes in the argument, participants might be less likely to take a stance, provide pieces of evidence, or persuade others. Instead of simply being civil, a political discussion should aim to encompass “conversation specifically aimed at political action” (Papacharissi, 2004, p. 281) and make comprehensive arguments validated with facts and data (Freelon, 2010; Halpern & Gibbs, 2013; Stroud et al., 2015; Theocharis et al., 2016). Chen (2017) speculated that both incivility and political deliberation can increase when message length constraints are relaxed; however, it is not known whether this effect would remain in the absence of the deindividuation differences between platforms. The study by

Oz et al. (2018) reported with observational and experimental analysis that political messages addressed to the U.S. White House on Twitter were significantly less likely to mention factual data, as compared to messages on Facebook.

Besides high-quality argumentation, political discussions also imply an exchange of arguments; consequently, participants are required not simply to speak, but also to listen and respond (Barber, 1984). Friess and Eilders (2015), for instance, emphasized the importance of interactivity as a key dimension of deliberative behavior. Other studies examining interactivity in deliberation, such as Habermas (1990), Walther et al. (2005), Stroud et al. (2015), and Himmelroos (2017), have stressed the importance of role-taking and empathy in a rational discussion.

To summarize, in general we anticipated that with more characters at their disposal, users could afford to be more civil, construct better arguments, and be more pleasant to each other in their political discussions. This is based on empirical findings from the many papers that have compared the quality of political deliberations across social networking sites with different affordances (Chen, 2017; Oz et al., 2018; Stroud et al., 2015), as well as previous studies that have theorized that improving opportunities to participate (e.g., designing social media platforms with better affordances) would lead to increases or improvements in deliberative behavior (Cook et al., 2007; Neblo, Esterling, Kennedy, Lazer, & Sokhey, 2010). We anticipated that extending the character limit might have afforded users more space to make a cogent argument or support their views with evidence, thus facilitating more sophisticated and deliberative political discussions (Chen, 2017; Theocharis et al., 2016). Accordingly, we posited our second hypothesis in two parts:

H2: Comments posted on Twitter after the 280-character limit change will comprise a better quality of (a) arguments and (b) interactions than before the change.

Data and methods

We built our data set of political discussions on Twitter by adopting the method of Theocharis et al. (2016) to identify the Twitter replies to 536 U.S. Congressmen and Congresswomen who were in office before 7 November 2017, when Twitter instituted a change in the character limit. Using the Twitter replies to members of Congress increased the odds that a particular tweet was part of a political discussion. We filtered a Twitter 1% sample, collected using the Twitter streaming Application Programming Interface (API), to retain all replies (i.e., tweets starting with politicians) to this set of Twitter handles occurring between 1 January 2017 and 31 March 2018. These dates gave us a larger pre-intervention and post-intervention observation period and the ability to assess whether our results remained stable if we considered different bandwidths in the regression equation.

In Table 1, we provide a summary of our data set with the number of tweets remaining after each pre-processing step. First, we performed language filtering

Table 1 Data Set Description

Number of observations	Replies in the 1% sample	After language filtering	After removing retweets	After removing early access
	2,101,856	1,869,481	398,278	358242
Number of Users	Unique Users	Non-compliers	Compliers	...
	204,902	62,180	39,246	...
Tweet Characteristics	Total Number of Words	Average Words per Tweet	Mean Daily Tweets	...
	7,668,123	21.40 (12.60)	772.56 (479.76)	...

Note: Standard errors are in parentheses.

to retain only English-language tweets. Next, we removed any tweets which were retweets and, thus, did not constitute an actual reply. We manually inspected the character limits of the tweets and found that, because of the differences in encoding, the actual lengths of tweets in the pre-intervention period was often up to 145 characters. Accordingly, we considered 145 characters as a better approximation to compliance than 140 characters. We removed tweets by the 1% of the population who were subjected to the intervention early (the early-access users) on 27 September 2017. Finally, our data set comprised 358,242 replies to U.S. politicians from January 2017 to March 2018.

Table 1 also provides statistics on the unique number of users in our data set. We also identified the compliers (users who tweeted more than 145 characters at least once after the character limit change) and the non-compliers (users who, despite the extension, never tweeted more than 145 characters). The Wilcoxon signed-rank test showed that compliers and non-compliers differed significantly across some of their account characteristics, such as the number of followers. Non-compliers were more likely to have been Twitter users for longer ($m = 4.43$ years), but compliers had a higher average number of tweets posted ($m = 31,681.93$, $SD = 52,822.50$) and a higher per-user occurrence in our data set ($m = 2.82$, $SD = 3.78$), as compared to non-compliers ($m = 29,408.74$, $SD = 52,030.67$ and $m = 2.11$, $SD = 2.13$, respectively), which suggests that compliers might have been more active on Twitter and already have been more engaged in politics than non-compliers. A comprehensive comparison of compliers and non-compliers is provided in the Supporting Information. Our within-subjects model specification controls for these individual differences and our results remained substantively similar to our other methods.

Operationalization

Based on previous work in analyzing political discourse, civility, and deliberation on social media, we identified a set of language features that can capture the quality of political discussions. Specifically, we focused on whether the tweets reflected uncivil or deliberative behavior. Furthermore, we distinguished the criteria followed to measure the quality of arguments: that is, the use of justification and constructiveness, from the criteria to measure the qualities of interactivity, as per [Friess and Eilders \(2015\)](#); i.e., reciprocity and empathy and respect). These are elaborated below. We have also described how they fit in with our hypotheses and research question in [Table 2](#).

1. We operationalized incivility in terms of the daily number of uncivil tweets or the average percentage proportion of uncivil words per tweet, as predicted by different machine learning and dictionary-based lexical methods.
2. We operationalized formality in terms of the decrease in the use of informal words or increase in politeness in political discussions.
3. We operationalized a measurement of deliberative quality in terms of the “justification” and the “constructiveness” attributes of political discussions, based on previous work that has used three criteria: the presence of supporting or external materials; the use of “comprehensive” arguments; and the use of numbers or statistics ([Oz et al., 2018](#); [Papacharissi, 2004](#); [Stroud et al., 2015](#)). Other work by [Esteve Del Valle, Sijtsma, and Stegeman \(2018\)](#) and [Rowe \(2015\)](#) has also identified the use of personal anecdotes, values, and ideologies to support claims. [Friess and Eilders \(2015\)](#) and [Steenbergen, Bächtiger, Spörndli, and Steiner \(2003\)](#) consider arguments to be constructive if, besides providing factual arguments, they also attempt to resolve conflicts or propose solutions.
 1. Justification reflects whether the author offers a justification, either based on personal experiences, values, and feelings or on data, links, and facts ([Esteve Del Valle et al., 2018](#); [Oz et al., 2018](#); [Rowe, 2015](#)).
 2. Constructiveness reflects the author’s attempt to move the conversation forward, build and bring about consensus, and resolve conflicts by pointing out facts, identifying common ground, or proposing solutions ([Friess & Eilders, 2015](#); [Steenbergen et al., 2003](#)).
4. Finally, we operationalized interactivity in two ways: reciprocity and empathy and respect. Reciprocity is defined in terms of writing that asks a genuine question and invites responses in the form of answers or further information ([Stromer-Galley, 2007](#); [Stroud et al., 2015](#)). We based our definitions on prior criteria to measure discursive quality ([Esteve Del Valle et al., 2018](#); [Steenbergen et al., 2003](#)) as positive comments that are sensitive or empathetic to others’ viewpoints and respectful towards other discussants.
 1. Reciprocity reflects whether the author engages with others in a substantial manner rather than a formal one, with a tweet intended to elicit a response or further information ([Friess & Eilders, 2015](#); [Rowe, 2015](#); [Stroud et al., 2015](#)).

2. Empathy and respect reflect the author's acknowledgment of or sensitivity to others, manifested in positive comments or an empathetic or a respectful response acknowledging other viewpoints (Esteve Del Valle et al., 2018; Steenbergen et al., 2003).

As a first step, we trained a supervised machine-learning classifier on 6,000 hand-annotated tweets, so that it could label each tweet in our data set according to whether it was relevant to politics. Labeling tweets as either being relevant to politics (1) or not (0) helped us to ensure that our inferences about political discussions were valid.

Uncivil behavior was measured in terms of the use of name-calling, profanity, hate speech, or the invocation of stereotypes of a homophobic, racist, sexist, or xenophobic nature (Chen, 2017). Communication scholars have argued that incivility is more than offensiveness. For example, Papacharissi (2004) defined incivility as impolite behavior that threatens democracy and has lasting repercussions on the common good. In other words, uncivil tweets must be offensive, but offensive tweets are not necessarily uncivil. Papacharissi's (2004) conceptualization was adopted by recent studies using human coding (e.g., Groshek & Cutino, 2016; Muddiman et al., 2018; Rowe, 2015; Santana, 2014). In this study, we applied and compared four different language-based models of incivility:

1. We trained an incivility classifier on a data set of 6,000 hand-annotated tweets from our data set, labeled as uncivil (1) or civil (0), and captured a wide range of abusive, racist, threatening, or exaggerative behavior common to online political incivility (Stroud et al., 2015).
2. We trained an offensiveness classifier on a data set of hand-annotated tweets provided by Davidson, Warmesley, Macy, and Weber (2017), which labels tweets as offensive (1) or inoffensive (0).
3. We applied the uncivil words dictionary, hand annotated by Muddiman et al. (2018) from a corpus of *New York Times* comments.
4. We applied the swear words dictionary from Linguistic Inquiry and Word Count (LIWC; Pennebaker, Booth, Boyd, & Francis, 2015a), which measures the percentage proportion of swear words in a text.

Additionally, some scholars have considered politeness and formality to be essential qualities for constructive political debate (Chen, 2017; Papacharissi, 2004). Politeness refers to adherence to etiquette and an extension of courtesy to fellow discussants (Papacharissi, 2004). We used the Stanford Politeness API to access a language model provided by Danescu-Niculescu-Mizil, Sudhof, Jurafsky, Leskovec, and Potts (2013) that scores each tweet on a scale of 0 to 1 for its politeness. Informality (or, conversely, the lack of formality) is used to reflect messages with less substance and is considered a less effective means of political discussion. We used the informality dictionary provided by LIWC (Pennebaker, Booth, et al., 2015), which defines informality as content that is relatively higher in its use of assents, fillers, swear words, and netspeak.

Table 2 The Deliberative Criteria Used to Measure the Quality of Political Discussion

Dimension	Operationalization	Source
Uncivil behavior		
Incivility	Whether a tweet contains abuses and vulgarity, threats, or exaggeration (1/0 binary coding)	This study
Offensiveness	Whether a tweet contains abuses and racial slurs (1/0 coding)	Davidson et al. (2017)
Uncivil words	The proportion of a tweet which comprises words from a curated list of offensive words (0 to 100 continuous values)	Muddiman, McGregor, & Stroud (2018)
Swear words	The proportion of a tweet which comprises words from a curated list of swear words (0 to 100 continuous values)	Linguistic inquiry and word count (Pennebaker, Booth, et al., 2015)
Deliberative behavior		
Politeness	Likelihood that the tweet is polite (0 to 100 continuous values)	Stanford's politeness API (Danescu-Niculescu-Mizil et al., 2013)
Informal words	The proportionate use of informal language, such as slang, fillers, swear words, and netspeak (0 to 100 continuous values)	Linguistic inquiry and word count (Pennebaker, Booth, et al., 2015)
Justification	Whether the tweet offers evidence in the form of feelings, experiences, or facts (1/0 binary coding)	This Study
Constructiveness	Whether the tweet offers a fact-check, search for common ground, or a solution (1/0 binary coding)	This Study
Reciprocity	Whether the tweet is intended to elicit an answer, information, or feedback (1/0 binary coding)	This Study
Empathy & respect	Whether the tweet contains empathy or respect for others (1/0 binary coding)	This Study

Note: API = Application Programming Interface.

Following the original definition by Gastil (2008), we sought to measure deliberative behavior in terms of the use of analytical and social language to build consensus, through facts and supporting rationales; raise and answer questions; consider and

rationalize alternatives; and express positions on issues. To our knowledge, there are no supervised machine learning classifiers available to label tweets according to their deliberative content automatically. Accordingly, we trained classifiers on a set of hand-annotated tweets and used them to label the entire data set.

Almost all of the operationalizations encode the presence or absence of a feature. Politeness is output as a continuous variable, constituting a likelihood percentage. The dictionary-based methods used to measure informality and uncivil words comprise normalized, within-tweet percentages, indicating the proportion of words that were representative of the category of interest. For example, a score of three for informality would imply that if a tweet comprised a hundred words, there would be three informal words.

Validation

For the incivility and the deliberative classifiers developed in this study, machine learning methods were implemented to train logistic regression classifiers on the language of 6,000 tweets, labeled by four annotators. The classifiers used a weighted function of the normalized frequency distribution of words and phrases in a tweet to assign a 0 or 1 label for the presence of incivility and each deliberative attribute. We validated the classifiers against held-out, labeled data in 10-fold cross-validation and obtained an average accuracy of 79% across all the categories. Details on the inter-coder agreement and the predictive performance are provided in Table 4 and 5 of the Supporting Information.

For measuring politeness and offensiveness, we used supervised machine learning models, trained on hand-annotated data developed by Danescu-Niculescu-Mizil et al. (2013) and Davidson et al. (2017). The politeness API has also been extensively used by the computer science research community (e.g., Althoff, Danescu-Niculescu-Mizil, & Jurafsky, 2014; Jongeling, Sarkar, Datta, & Serebrenik, 2017; Tan, Niculae, Danescu-Niculescu-Mizil, & Lee, 2016) in linguistic analyses of social media text. The classifier and the annotated data set on offensiveness have been validated in subsequent studies (Almeida, Souza, Nakamura, & Nakamura, 2017; Olteanu, Talamadupula, & Varshney, 2017).

For measuring swear words and informality, we used the dictionaries provided by LIWC 2015 (Pennebaker, Boyd, Jordan, & Blackburn, 2015b), a computerized program developed by psychologists to automatically categorize words in a text (Pennebaker, Booth, et al., 2015). Dictionaries of LIWC have been validated in subsequent language analyses of social media posts.¹ Further details about the supervised and unsupervised methods used to mine these features are provided in the Supporting Information.

Interrupted time series regression model

Our primary approach was an interrupted time series analysis (ITS) to determine whether the character-limit change induced an improvement in political discussions.

ITS is a variation of regression discontinuity designs where the running variable is time (Lopez Bernal, Cummins, & Gasparrini, 2017). This approach was ideal because Twitter data is timestamped, with a high frequency of daily measurements and a well-defined moment of intervention. ITS design requires a clear differentiation of the pre-intervention and post-intervention period (Lopez Bernal et al., 2017): in our case, the extension of the character limit from 140 to 280 on 7 November 2017. The unit of analysis in this set of regressions was the daily mean score of an uncivil or a deliberative attribute. The quantity of interest was the immediate change in the different attributes measuring uncivil and deliberative behavior after 7 November 2017. To tackle the inconsistency of the numbers of tweets on each day and avoid Type I errors, we bootstrapped the analysis for 100 iterations, sampling the approximate daily mean of 700 tweets per day for each analysis, and reported the average effect sizes and standard errors across all the iterations. All variables were scaled to values between 0 to 100 and mean-centered before analysis, in order to more readily interpret the interaction term. The independent variable was dichotomous, indicating whether it was before or after the intervention. The number of days since the intervention is represented by T in the following ordinary least squares (OLS) model:

$$Y_{feature,t} = \beta_0 + \beta_1 T + \beta_2 X_t + \beta_3 TX_t$$

In the above model, T is the relative time distance from 7 November 2017. For example, time equals -3 on 4 November 2017 and equals 7 on 14 November 2017. X is a dummy variable indicating whether the tweets were published before (coded 0) or after the intervention (coded 1). Therefore, β_2 indicates the intercept shift following the character limit intervention on the feature value, while β_3 shows the slope change after the character limit intervention. Including quadratic or cubic time trends (described in the Supporting Information) did not substantively change our results.

Although we used the standard ITS model, we focused on β_2 (i.e., the intercept shift) and not β_3 (i.e., the slope shift). The slope, in part, indicated whether any post-treatment intercept shift returned to baseline after some time. This reversion to the mean may even be unrelated to the character limit change. The OLS estimates for the slope change are reported in Table 6 of the Supporting Information.

Using this regression model, we examined the effect of the character limit intervention on various uncivil and deliberative attributes ($Y_{feature}$). The regression results in Table 3 show the average of the 100 iterations, while the locally estimated scatterplot smoothing (LOESS) figures show one random iteration for the purposes of illustration. We focused on the 100 days before and 100 days after the character switch (i.e., bandwidth set to 100). For expository purposes, we also present a simple difference in means in the feature prominence pre- and post-intervention.

Additional specifications

Effects among compliers

Using the ITS framework, we also compared the pre-intervention levels among the entire sample to post-intervention levels only among tweets that contained more than 145 characters (compliers) and among those that contained less than 145 characters (non-compliers). We expected that the intervention would have a significant effect on the uncivil and deliberative characteristics of the tweets by the compliers.

Instrumental variable estimates

Since not every person complied with the treatment (used more than 145 characters), we also used a fuzzy regression discontinuity framework that considered the intervention day as a discontinuity and time as the running variable (Hausman & Rapson, 2018). The intervention exogenously increased the probability that someone used more than 145 characters. Our causal estimate in this model is the Local Average Treatment Effect: that is, the effect among those who used more than 145 characters. A dummy indicating whether the tweet was posted before (coded 0) or after (coded 1) 7 November was the instrumental variable. The number of characters in a tweet was the endogenous variable (coded 0 if less than 145 characters and 1 if more than 145 characters).

Within-subject effects

Since some people who are more likely to tweet different types of messages may be more likely to use more than 145 characters, we also tested our results for any evidence of self-selection bias. We limited the data set to those subjects that tweeted before and after the intervention and included subject fixed effects, which effectively examined the effect of the character limit change within subjects, washing away time-invariant effects.

Bandwidth effects

To test whether our results were an artifact of the bandwidth chosen in our ITS models, we also replicated our ITS models with different bandwidths of data.

Placebo effects

We replicated our analysis on a data set of tweets posted in 2016, using the same data collection method as above, to identify 66,927 replies to the same U.S. politicians in the 100-day bandwidth before and after 7 November 2016. We replicated our experimental analysis to calculate the difference in means and a linear model specification. We report the average standardized coefficients and standard errors for 100 iterations, conducted on an average of 400 daily observations, sampled with replacement from the data set.

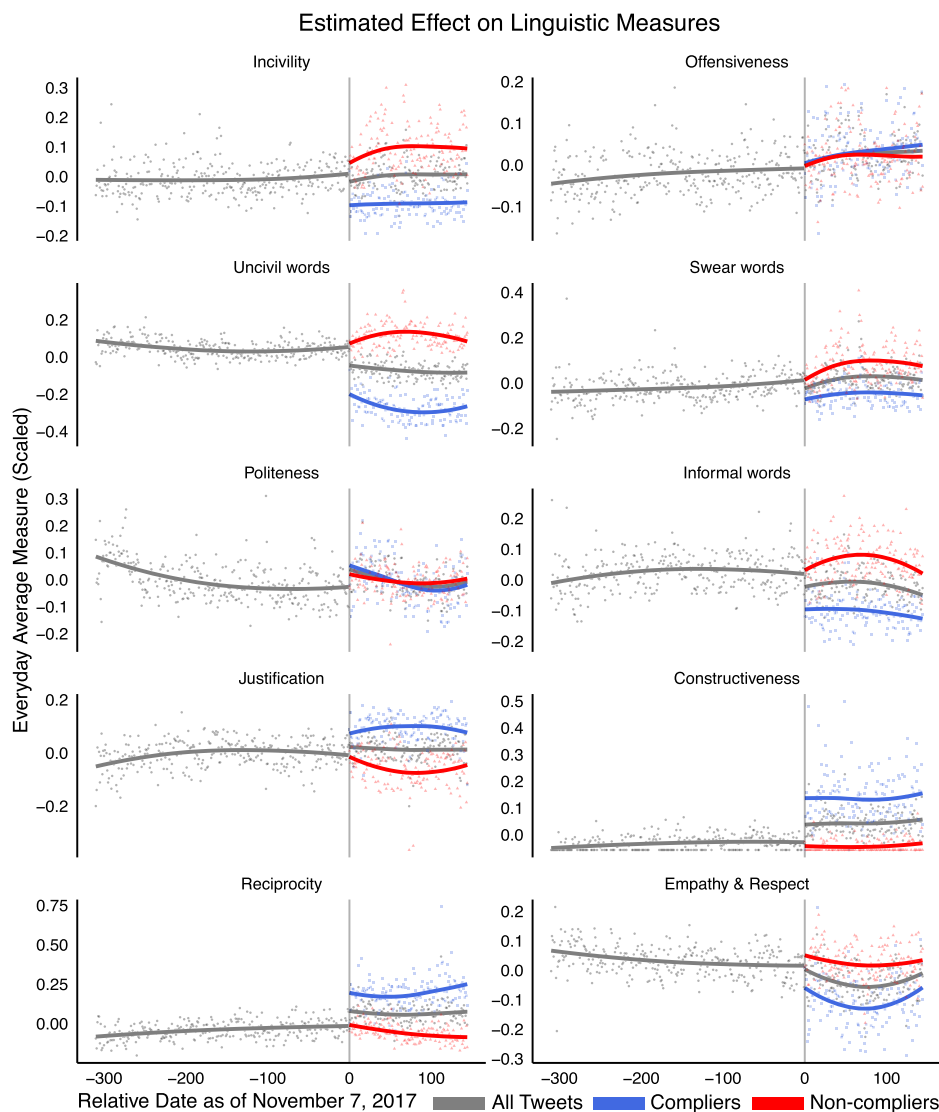


Figure 1 Daily trends of uncivil and deliberative behavior from January 2017 to March 2018 for political discussions on Twitter. The solid curve is generated by locally weighted (LOESS, locally estimated scatterplot smoothing) regression of the uncivil and deliberative attributes on sequential day numbers, with no adjustment for covariates. Data were aggregated as day-level means here to facilitate visualization.

Results

As a manipulation check, we first assessed whether the intervention affected the number of words used per tweet. The average number of words per tweet increased

significantly after the character limit intervention ($\beta .71; p = 3.4e^{-5}$). This effect did not decline over time (coefficient of interaction $.003; p = .17$).

The core analysis was performed on 358,242 tweets, of which 146,878 were posted after the intervention (Twitter's character limit change). We found that 99% of the annotated tweets were relevant to politics, which gives us confidence that our findings apply to describe online political discussions. The distribution of different uncivil and deliberative attributes and their intercorrelation is provided in Table 1 and Figure 1 of the Supporting Information. Compliers posted 70,440 tweets (48.0%) after the intervention. Figure 1 shows the change in the uncivil and deliberative attributes in the period leading up to and after the intervention, where the solid curves were generated by locally weighted regression of the attributes on sequential day numbers, with no adjustment for covariates. Post-intervention, the blue curve reflects the trend for compliers, while the red curve depicts the non-compliers.

This visualization provides the first set of clues regarding the effect of the intervention on online political discourse. Post-intervention, compliers (blue) and non-compliers (red) appeared to be markedly different across most of the uncivil and deliberative characteristics. No discontinuities at the intervention were observed for offensiveness. Incivility, swear words, and informal words showed a discontinuous decrease on the day of the intervention. Politeness, justification, constructiveness, and reciprocity showed a discontinuous increase, and empathy and respect decreased discontinuously.² In considering the slopes of these curves (the β_3 coefficients, reported in Table 6 of the Supporting Information), the findings suggest that politeness showed a temporary level change and would revert to the pre-intervention mean over time. On the other hand, the decrease in the use of uncivil words and empathy and respect reflect both a level and a slope change, and their presence in tweets is likely to further decrease over time.

We formally tested these differences in Table 3. These OLS estimates are fit to the daily trends for a bandwidth of 100 days. Aggregating by day, Column 1 presents an estimate of the mean difference in the attributes before and after the intervention. Column 2 fits a linear function to the data on either side of the intervention date. Columns 3 and 4 provide the results, considering all the tweets pre-intervention and separating the effects among compliers and non-compliers post-intervention. Column 5 provides the local average treatment effects among compliers, following a fuzzy regression discontinuity design implementation. Columns 6 and 7 report the estimates after considering subject fixed effects, for those in our data set who tweeted both pre- and post-intervention.³

To answer our research question, we observed that two out of four measures of uncivil behavior significantly changed post-intervention across all specifications, to make the discussions more civil overall. Offensiveness only differed in the first (and least credible) specification. Average treatment effects reporting reduced uncivil behavior among compliers are significant in three of the four specifications. Among compliers, reduced uncivil behavior was significant in three of four specifications: lower incivility (Column 3: $\beta = -.11, p = 3.86e^{-8}$; Column 5: $\beta = -.12, p = 2.73e^{-12}$),

Table 3 Ordinary Least squares Estimates of the Effect of Platform Change With Different Model and Treatment Specifications

1	2	3	4	5	6	7
Difference in means	Linear	Linear compliers	Linear non-compliers	IV estimates	Within-subject: all	Within-subject: compliers
Uncivil behavior						
Incivility	.00 (.009)	-.11** (.015)	.05* (.019)	-.12** (.017)	-.00 (.017)	-.03 (.017)
Offensiveness	.03** (.009)	.014 (.019)	.014 (.019)	.02 (.016)	.03 (.018)	.05* (.019)
Uncivil words	-.11** (.008)	-.27** (.018)	.04* (.018)	-.23** (.015)	-.10** (.024)	-.21** (.021)
Swear words	.02 (.009)	-.07** (.016)	.02 (.021)	-.10** (.017)	-.01 (.017)	-.01 (.017)
Deliberative behavior						
Politeness	.04** (.009)	.08** (.021)	.05* (.019)	.11** (.017)	.04* (.020)	.04 (.021)
Informal words	-.04** (.009)	-.12** (.018)	.03 (.021)	-.13** (.017)	-.05** (.019)	-.12** (.018)
Justification	.01 (.008)	.08** (.016)	-.04** (.020)	.06** (.016)	.00 (.020)	.04 (.022)
Constructiveness	.07** (.007)	.16** (.024)	-.02 (.021)	.13** (.015)	.05 (.024)	.14** (.024)
Reciprocity	.08** (.009)	.18** (.023)	.00 (.017)	.19** (.018)	.09*** (.022)	.19** (.022)
Empathy & respect	-.06** (.009)	-.08** (.011)	.03 (.018)	-.04* (.017)	-.04 (.022)	-.07** (.021)
Observations	200	200	200	180634	15271	7382

Note: Bandwidth = 100 days. Standard errors are shown in parentheses. * $p < .05$. ** $p < .001$.

lower proportions of uncivil words (Column 3: $\beta = -.27, p = 7.6e^{-32}$; Column 5: $\beta = -.23, p = 4.05e^{-54}$; Column 7: $\beta = -.21, p = 1.20e^{-15}$) and swear words (Column 3: $\beta = -.07, p = 1.47e^{-5}$; Column 5: $\beta = -.10, p = 6.92e^{-10}$). However, incivility and the use of swear words did not significantly change in the within-subject specification among compliers.

To test H1, we considered the results for informality. Compliers were less likely to use informal language after the character limit change (Column 3: $\beta = -.12, p = 1.49e^{-8}$; Column 5: $\beta = -.13, p = 3.08e^{-15}$; Column 7: $\beta = -.12, p = 5.98e^{-11}$). Politeness significantly increased among compliers (Column 3: $\beta = .08, p = 4.0e^{-4}$; Column 5: $\beta = .11, p = 3.86e^{-11}$) and was marginally significant at the within-subject level (Column 7: $\beta = .04, p = .07$). While politeness also increased among non-compliers, the effect was 33% larger among the compliers than non-compliers. In summary, our findings show that, with the increase in the character limit per post, political discussions on Twitter became more civil. Therefore, H1 is supported by the results.

To test H2a, we considered the results for justification and constructiveness. Across all model specifications, compliers were more likely to use justifications after the character limit change (Column 3: $\beta = .08, p = 1.11e^{-4}$; Column 5: $\beta = .06, p = 3.36e^{-4}$), and this result was marginally significant at the within-subject level (Column 7: $\beta = .04, p = .08$). They were also more likely to write constructive tweets across all the specifications (Column 3: $\beta = .16, p = 1.86e^{-9}$; Column 5: $\beta = .13, p = 2.32e^{-16}$; Column 7: $\beta = .14, p = 1.5e^{-6}$) in political discussions after the intervention.

To test H2b, we considered the results for reciprocity and for empathy and respect. Across all model specifications, compliers were more likely to manifest reciprocity in their political discussions after the character limit change (Column 3: $\beta = .18, p = 6.08e^{-11}$; Column 5: $\beta = .19, p = 4.209e^{-28}$; Column 7: $\beta = .19, p = 1.12e^{-16}$). However, they were also less likely to show empathy and respect (Column 3: $\beta = -.08, p = 5.2e^{-4}$; Column 5: $\beta = -.04, p = .02$; Column 7: $\beta = -.07, p = 6.7e^{-4}$). Thus, while interpersonal engagement increased, there was a decrease in the general empathy and respect shown towards others.

Taking together the results from H2a and H2b, we can say that the results partially support H2. In terms of the analytical processes of political discussions, there was a promising increase in the use of facts and data to support arguments. At the same time, however, the quality of socializing appears to have decreased, with writing becoming more formal yet less sensitive to other viewpoints.

Interpretation

We noted that the post-intervention differences for political content are small at the tweet level and the subject level (Columns 5 and 7 of Table 3). To interpret the effect of the character limit change on the political discussions, we examined the effects when the data were mean-centered at the day level and found that, in many cases, the

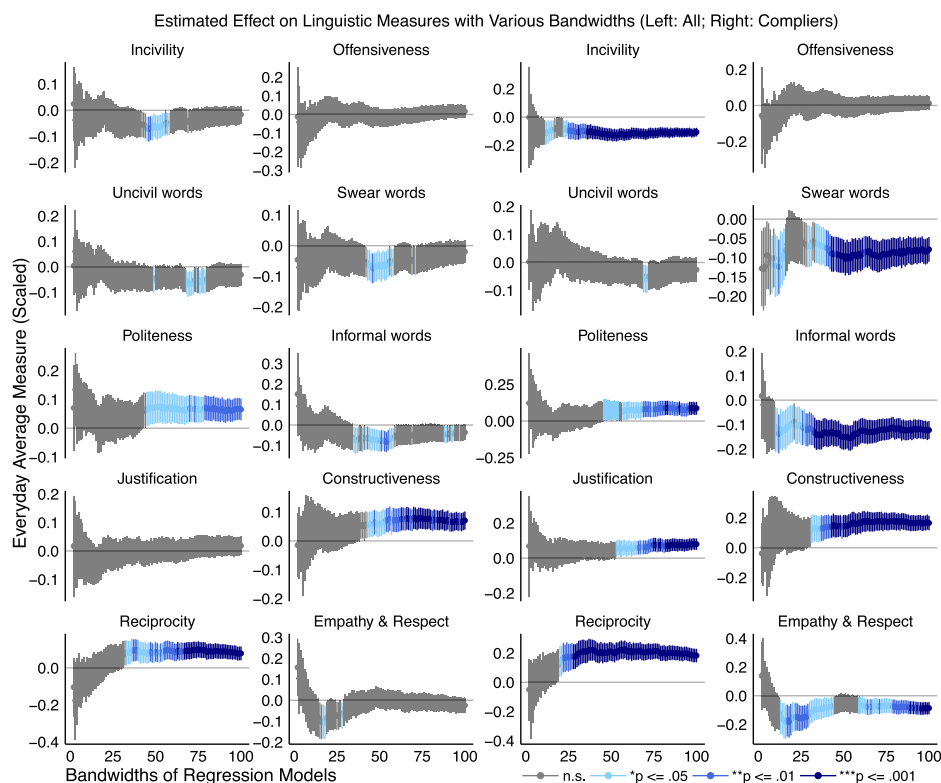


Figure 2 Estimates of the average change in uncivil and deliberative attributes after the intervention by fitting a linear model over (a) all observations ($N = 358,242$) and (b) compliers ($N = 281,804$), using various bandwidths $[0, 100]$, as plotted on the x-axis. Vertical lines denote 95% confidence intervals for each estimate.

day-level shift among compliers was as high as 1.3%.⁴ Consequently, we expect that, across millions of tweets and people, small within-subject effects can lead to a large difference in the overall characteristics of the political conversation, as visualized in the discontinuities in Figure 1.

Robustness and placebo tests

Figure 2 displays the estimated treatment effects using the linear functional form, with various bandwidths ($N = [1, 100]$). The left panel shows the result by fitting the linear model to all observations, while the right panel considers only compliers post-treatment. Shaded points show significant estimates at three different levels ($p \leq .05$, $p \leq .01$, or $p \leq .001$). These trends show that the choice of bandwidth affects the significance of effects when there is a smaller sample size. Importantly, our findings do not appear to be contingent on the choice of bandwidth because, in most of the cases,

the direction of the effect remained consistent (i.e., either all effects were negative or all effects were positive) across large and small bandwidths.

For the placebo data set collected in November 2016, when there was no real intervention or increase in the Twitter character limit, we observed that the results did not repeat the main trends of Table 3 following the placebo treatment. These results are reported in Table 7 of the Supporting Information.

Discussion and recommendations

Many have argued that popular social media platforms might not be suitable for constructive discussions, citing the negative human behavior demonstrated online, such as toxicity, incivility, and a lack of empathy (e.g., Anderson, Brossard, Scheufele, Xenos, & Ladwig, 2014; Baek, Wojcieszak, & Delli Carpini, 2012; Coe, Kenski, & Rains, 2014). Some have suggested that a solution to online incivility lies in exposing people to different content or to exhortations to act civil (e.g., Kim, 2015; Munger, 2017). We explored an alternative solution by examining the impact of a technological affordance change on political discussions. While Twitter's implementation of a character limit change to double the length of tweets led to less uncivil political discussions and more deliberative political discussions, it also decreased empathy and respect among the discussants. These results highlight the importance of the careful design of discussion forums for the quality of political discussions.

Our study highlights the potential impact of affordances on online political communication in general. Firstly, it hints at how affordances can trigger a change in media effects, because we observed an increase in information-sharing behavior (justification and constructiveness; precisely, the sharing of facts) after the change of affordances. Another implication is the impact of affordances on the perceived credibility of online messages. It would be interesting to consider how a change in other affordances—for example, in anonymity, the visibility of social cues, or the placement of action buttons (Matias, 2019; Stroud et al., 2015)—could change online social norms or facilitate conversations across political lines of difference (Settle & Carlson, 2019).

Secondly, the findings suggest that a change in affordances can also trigger a change in online social relationships. While the character limit improved overall civility in political discussions, the decrease in empathy and respect among compliers is a cause for concern. That is to say, while people were more polite after the character limit change, they were less likely to be empathetic or respectful to other people's comments. The findings offer a different dimension to measure the tone of deliberation. Previous studies have implied that less respectful messages may be considered more persuasive (Fridkin & Kenney, 2008) and more authentic (Benson, 2011), perhaps signaling sincerity and the speaker's high stakes in the disagreement (Benson, 2011). Future work could explore whether changing interactivity affordances could help to include more voices, through friendlier responses, or change the perceptions of authenticity and trust in online political discussions.

Finally, there is cause for concern that a change in affordances creates new social norms and structural inequalities in the online public sphere. Our findings show that the effects of Twitter's new affordance were largely limited to those who self-selected into applying them in their daily communication. Therefore, the overall improvement in the argumentative quality of political discussions on Twitter may have only mattered for those social media users who were already politically engaged. Scholars have shown that elite users—already more engaged, articulate, and authoritative—lead the political rhetoric in the dominant, rather than the public, sphere of debate (e.g., Fraser, 1990; McGregor, 2018; Papacharissi, 2004).

As far as we know, no previous study has coded the deliberative characteristics of a data set of this size, either manually or automatically; thus, this study offers the first benchmark of the deliberative characteristics of everyday political discussions. It revealed that over 99% of all messages were labeled as relevant to politics, which validates our and Theocharis et al.'s (2016) sampling method, and also belies the general perception of these discussions as mere chatter. Secondly, only about 4% of daily political messages were predicted to be uncivil, which again contradicts the general perception of online political discussions as only toxic and frustrating.

As with all quasi-experimental designs, we cannot discount all threats to internal validity (Shadish, Cook, & Campbell, 2002). In particular, a threat to internal validity due to historical events remains possible. It is notable that 8 November 2017 was the first anniversary of the 2016 United States presidential election, which was very close to the date of the Twitter character limit change. The event might have briefly changed the uncivil and deliberative characteristics of the tweets around that time. However, the effect would have eventually faded away, while the new 280-character limit would not. Moreover, as our analysis shows, the effect of the character limit change remained months after the event. Furthermore, a placebo test conducted using the data of November 2016 strengthens our claims.

The findings of our study have practical implications for future research. First, our study highlights the need to conduct more time series analyses with social media data to better understand evolving social norms on social media platforms, and their impact on the public sphere. Given the nature of our sample, our findings explain the change in individual tweets; however, inferences at the conversation level would require all the tweets and replies to be in a single thread. Although we employed the Twitter streaming API in our study, which only returns a 1% random sample of all public tweets, we recommend scholars use Twitter's firehose stream when possible, which charges based on the date range and the number of tweets requested by researchers. Studies have debated the sampling validity of the streaming API for quite some time (for details, see González-Bailón, Wang, Rivero, Borge-Holthoefer, & Moreno, 2014; Morstatter, Pfeffer, & Liu, 2014; Morstatter, Pfeffer, Liu, & Carley, 2013). However, a time-series analysis requires a wide band of data, which often makes it monetarily unrealistic to use the full-access, yet expensive, alternative.

While this study offers a starting point, much needs to be done to examine whether these findings would lead to changes in overall political engagement, or

generalize to other political or language norms. Political engagement on social media is especially important in countries with a greater reliance on Internet election campaigning (Ahmed & Skoric, 2014) and in countries with restricted press freedom or a more authoritarian government (Ahmed & Cho, 2019). In terms of generalizability to other cultures and language norms, we should note that, as per the original conceptualization, an affordance denotes the relationship between the environment and the observer (Gibson, 1977). Twitter noted that only 0.4% of Japanese tweets hit the 140-character limit, whereas 9% of English tweets do. In cross- or multi-lingual contexts, there is often more than a character limit at play, and even a universal character limit actually has rather different effects in different languages (Liao, Fu, & Hale, 2015). We thus encourage other researchers to extend our research and examine the impact of affordances on online social norms in other environments and languages.

Conclusion

Through this study, we have shown how the design decisions made by computer engineers and technology companies might have a profound impact on democratic processes (Forestal, 2017). In light of recent events where communication technology had a detrimental effect on democracy, such as the compromise of personal data around the 2016 U.S. presidential election (Cadwalladr & Graham-Harrison, 2018), rampant misinformation on social media (Del Vicario et al., 2016), and some evidence of online echo chambers (at least on social media; e.g., Barberá, Jost, Nagler, Tucker, & Bonneau, 2015; Flaxman, Goel, & Rao, 2016; Sunstein, 2017), we contend that technology companies have a responsibility to make communication platforms more friendly for political discussions. However, this too should be implemented with careful consideration of users' rights; this study does not advocate technological determinism as a catch-all solution for improving the online political sphere.

The increasing accessibility of massive data sets and the development of new tools for computational and linguistic analyses have enabled scholars to use more sophisticated techniques to model computer-mediated communication; this also has broader implications for the study of political communication. Methodological frameworks to identify causal effects, such as instrumental variables and regression discontinuity designs, are also becoming increasingly popular in social science disciplines (e.g., Dunning, 2008). We hope that this study encourages others in communication to combine computational approaches with the tools of causal inference to identify media effects in the digital world.

To summarize, while an online political utopia may remain a pipe dream, this study has shown that we can make the current environment more hospitable to democratic discourse. While platforms are understandably cautious when it comes to censoring or promoting certain content or users, this study indicates that they have other tools that will promote healthy political discourse. Internet platforms are always A/B testing their products to increase profits and revenue and to improve

user engagement and satisfaction (Hindman, 2018). They may also want to test the impact of such design changes on discussions' health. Of course, platforms will need to balance their profit motives for their public interest motives when making these decisions, but we hope that the latter will play a significant role in this calculus.

Supporting Information

Additional Supporting Information may be found in the online version of this article.

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Notes

¹ See Tausczik and Pennebaker (2010) for more details about validation and Pennebaker, Booth, et al. (2015) for details on LIWC's construction and inter-coder reliability.

² Incivility solely measures the presence of abuses, extremist language, or vulgarity. Justification solely comprises evidence of facts, data, or links in the tweets, and constructiveness solely comprises the fact-checking aspect. Although in the annotation task, tweets were labeled for the presence or absence of the individual facets of incivility, justification, and constructiveness, respectively, there were not enough positive instances of other aspects to warrant training a classifier.

³ A post hoc analysis revealed that, for a margin of error of 4% in our within-subject analysis and given the number of subjects ($N = 7382$) and the smallest effect size (0.05) in the present study, we obtained a statistical power of 0.99.

⁴ The results are reported in Table 6 of the Supporting Information.

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